QingPeng Lam

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Education

University of Michigan — Ann Arbor, MI
Bachelor of Science in Computer Science, Minor in UX Design | GPA: 3.8 — *Aug. 2024 – Dec. 2026*

Experience

Software Engineering Intern — May 2026 – Aug. 2026
USAA — *San Antonio, TX*

- Built **12 reusable C# AI developer tools and MCP integrations** that orchestrated production-log retrieval, root-cause analysis, code-fix generation, unit testing, Git operations, and merge-request creation across **22 workflows and 6 internal projects**.
- Diagnosed **120+ application and CI/CD failures** by tracing distributed service interactions, source-code errors, production logs, and GitLab pipelines across **Docker and Podman** environments.
- Engineered a **full-stack React enterprise-search platform** integrating frontend components, backend services, and indexing pipelines to unify retrieval across internal and external data sources.

Data Systems Engineering Intern — Aug. 2025 – May 2026
University of Michigan Institute for Social Research — *Ann Arbor, MI*

- Engineered **Python and SQL ETL pipelines** with validation, deduplication, and transformation stages to convert raw research data into reliable datasets for recurring analysis and reporting.
- Built **REST APIs and interactive dashboards** that transformed research datasets into reusable metrics and reporting tools for **200+ researchers and program stakeholders**.
- Automated data preparation, report generation, bibliography processing, and documentation workflows with **Python**, replacing repetitive manual processes with reproducible pipelines.

Software Engineer Intern — Aug. 2024 – June 2025
Michigan Hillel — *Ann Arbor, MI*

- Built **Python, pandas, NumPy, and SQL ETL pipelines** to consolidate, clean, and validate **3,000+ records** from multiple operational data sources, **reducing reporting errors by 35%**.
- Developed **JavaScript and Tableau dashboards** backed by SQL data models and third-party APIs to centralize engagement and operational metrics for recurring planning and reporting workflows.

Projects

Mobile LLM Harness | *SwiftUI, Python, Starlette, SQLite, AWS EC2, HTTP/2, SSE* — Sep. 2026 – Present

- Engineered an **end-to-end SwiftUI LLM client and Python backend** with **5 API endpoints**, SSE token/reasoning streaming, persistent conversation history, and runtime switching across **5 LLMs**, supporting **500+ conversation turns**.
- Deployed a **TLS-secured harness on Ubuntu AWS EC2** with async HTTP/2 request handling, SQLite persistence, and remote model serving; validated **20 concurrent LLM requests with zero failures**.

DevDoctor | *Python, FastAPI, PostgreSQL, Ray Serve, MLflow, Docker* — June 2026 – Present

- Architected and built a **multi-tenant platform** that converts Sentry, GitHub, and GitLab failures into clustered incidents, AI-generated diagnoses, sandbox-validated patches, human approvals, and automated Draft PRs/MRs.
- Productionized the ML lifecycle with **MLflow model versioning**, **Ray Serve autoscaling**, PostgreSQL row-level security, OIDC/SCIM access control, durable workers, encrypted recovery, and **260+ automated tests**.

Technical Skills

Languages: Python, C++, Java, JavaScript, C#, SQL, HTML/CSS
Software Engineering: REST APIs, Data Structures & Algorithms, Web Application Development, Unix/Linux, Git, GitHub, GitLab CI/CD, Unit Testing
Distributed Systems & Infrastructure: AWS,Distributed Systems, Docker, Podman, PostgreSQL, Ray Serve
AI/ML & Information Retrieval: AI Productivity Tools, AI Agents, MCP, Generative AI, Machine Learning, Information Retrieval, Vector Search, Embeddings, MLflow, PyTorch, TensorFlow, Transformers.js
Frameworks & Libraries: FastAPI, React, Node.js, Express, pandas, NumPy